

Rationalizable strategies via iterated elimination of never-best responses

Consider the following game. Using iterated elimination of never-best responses, find the set of rationalizable strategies. Consider only pure strategies and best responses to pure strategies of the opponent.

	C_1	C_2	C_3	C_4
R_1	(10, 7)	(8, 8)	(0, 6)	(2, 6)
R_2	(6, 5)	(2, 3)	(5, 1)	(7, 4)
R_3	(0, 4)	(5, 8)	(3, 7)	(5, 10)
R_4	(4, 6)	(9, 8)	(6, 9)	(1, 1)

Solution

	C_1	C_2	C_3	C_4
R_1	(<u>10</u> , 7)	(8, <u>8</u>)	(0, 6)	(2, 6)
R_2	(6, <u>5</u>)	(2, 3)	(5, 1)	(<u>7</u> , 4)
R_3	(0, 4)	(5, 8)	(3, 7)	(5, <u>10</u>)
R_4	(4, 6)	(9, 8)	(6, <u>9</u>) [*]	(1, 1)

Teal underlines indicate the row player's best responses, and red underlines indicate the column player's best responses. The asterisk marks a Nash equilibrium, where both players are best responding to each other.

The unique pure-strategy Nash equilibrium is (R_4, C_3) , with payoffs (6, 9). Strategies played in a pure-strategy Nash equilibrium survive every round of iterated elimination of never-best responses: each remains a best response to the opponent's equilibrium strategy. Other strategies may also survive in general, although in this game only R_4 and C_3 remain.

A strategy is a never-best response if it is not a best response to any remaining pure strategy of the opponent. At each round, eliminate such strategies and recompute best responses in the reduced game.

1. First round.

For the row player, maximize the first payoff within each column:

$$BR_1(C_1) = \{R_1\} \quad BR_1(C_2) = \{R_4\} \quad BR_1(C_3) = \{R_4\} \quad BR_1(C_4) = \{R_2\}$$

Thus, R_3 is never a best response.

For the column player, maximize the second payoff within each row:

$$BR_2(R_1) = \{C_2\} \quad BR_2(R_2) = \{C_1\} \quad BR_2(R_3) = \{C_4\} \quad BR_2(R_4) = \{C_3\}$$

Every column is a best response to some row.

Therefore, eliminate R_3 only.

2. Second round.

The reduced game is

	C_1	C_2	C_3	C_4
R_1	(10, 7)	(8, 8)	(0, 6)	(2, 6)
R_2	(6, 5)	(2, 3)	(5, 1)	(7, 4)
R_4	(4, 6)	(9, 8)	(6, 9)	(1, 1)

Each remaining row is still a best response to some column. However,

$$BR_2(R_1) = \{C_2\} \quad BR_2(R_2) = \{C_1\} \quad BR_2(R_4) = \{C_3\}$$

Column C_4 was a best response only to the eliminated row R_3 .

Therefore, eliminate C_4 .

3. Third round.

The reduced game becomes

	C_1	C_2	C_3
R_1	(10, 7)	(8, 8)	(0, 6)
R_2	(6, 5)	(2, 3)	(5, 1)
R_4	(4, 6)	(9, 8)	(6, 9)

The row player's best responses are

$$BR_1(C_1) = \{R_1\} \quad BR_1(C_2) = \{R_4\} \quad BR_1(C_3) = \{R_4\}$$

All remaining columns are still best responses to some remaining row.

Therefore, eliminate R_2 , which is no longer a best response to any remaining column.

4. Fourth round.

With only R_1 and R_4 , the game is

	C_1	C_2	C_3
R_1	(10, 7)	(8, 8)	(0, 6)
R_4	(4, 6)	(9, 8)	(6, 9)

Both rows remain best responses, whereas

$$BR_2(R_1) = \{C_2\} \quad BR_2(R_4) = \{C_3\}$$

Therefore, eliminate C_1 .

5. Fifth round and termination.

The remaining game is

	C_2	C_3
R_1	(8, 8)	(0, 6)
R_4	(9, 8)	(6, 9)

Since $9 > 8$ and $6 > 0$, R_4 is the unique best response to both remaining columns.

Therefore, eliminate R_1 .

Against the only remaining row R_4 , the column player prefers C_3 to C_2 , since $9 > 8$.

Therefore, eliminate C_2 . The surviving strategy sets are

$$S_1^\infty = \{R_4\} \quad S_2^\infty = \{C_3\}$$

Under the stipulated restriction to pure-strategy best responses, the unique rationalizable profile is (R_4, C_3) , with payoffs (6, 9).